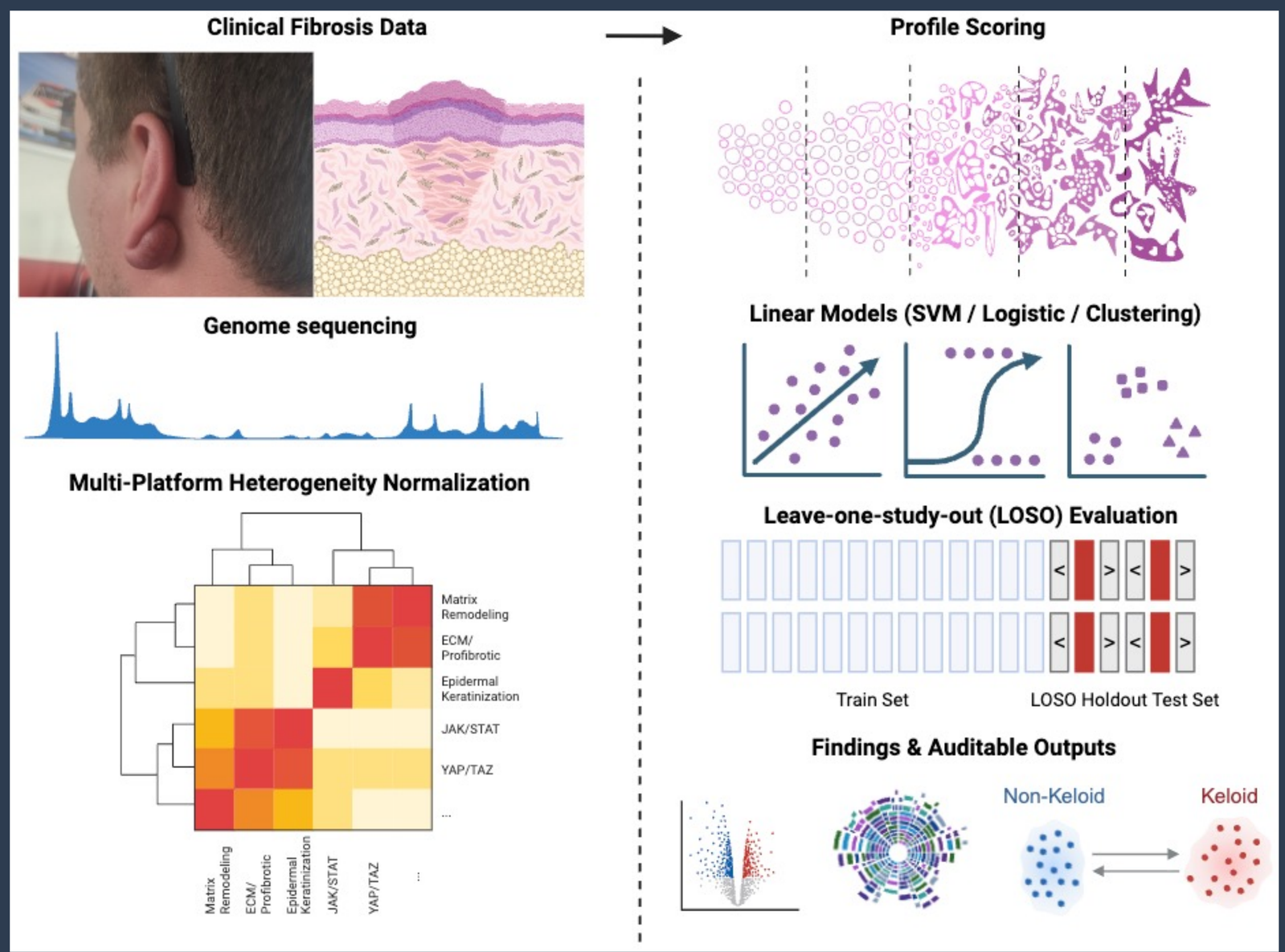
# KeloidBench: Interpretable machine learning on fibroblast program features for keloid detection

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## Introduction

- Keloids are pathological scars that extend beyond the original wound and can cause pain, itching, and recurrence after treatment.
- Molecular studies of keloids exist, but **public transcriptomic datasets are small, heterogeneous, and inconsistently labeled** (tissue definition, collection site/time, platforms, and preprocessing).
- Reported **marker genes and hub-gene signatures often fail to replicate** across cohorts due to label noise and domain shift.
- Progress needs a leakage-safe, cross-study benchmark that reflects real generalization (study-held-out evaluation rather than random splits).
- We propose KeloidBench, an interpretable ML pipeline that compresses gene expression into fibroblast activation program scores and produces calibrated probabilities + abstention for ambiguous cases.
- Goal: enable public-data triage (prioritize hypotheses, flag questionable samples, and justify matched-cohort/biobank follow-up) rather than just clinical diagnosis. Clinicians can upload expression files / GEO accession datasets with automated matrix normalization/scoring and prediction via this model.




## Methods & Product Architecture

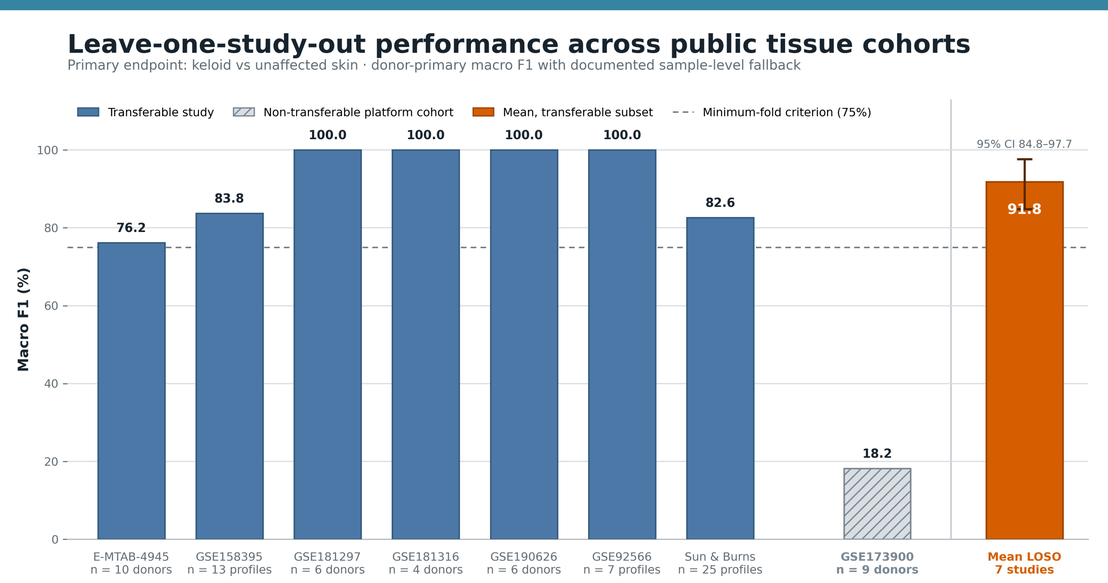
- Harmonizes public tissue-expression cohorts to a common gene vocabulary and keloid-versus-unaffected-skin endpoint.
- Converts expression values into within-sample gene ranks to reduce platform and normalization differences.
- Aggregates ranked genes into fibroblast, fibrosis, immune, keratinocyte, and endothelial program scores.
- Uses nested leave-one-study-out validation, keeping each test study isolated from feature, model, and threshold selection.
- Combines the three best training-study configurations through ensemble voting to predict the unseen cohort.
- Applies a fixed confidence policy that abstains on uncertain samples and reports program-level feature contributions.

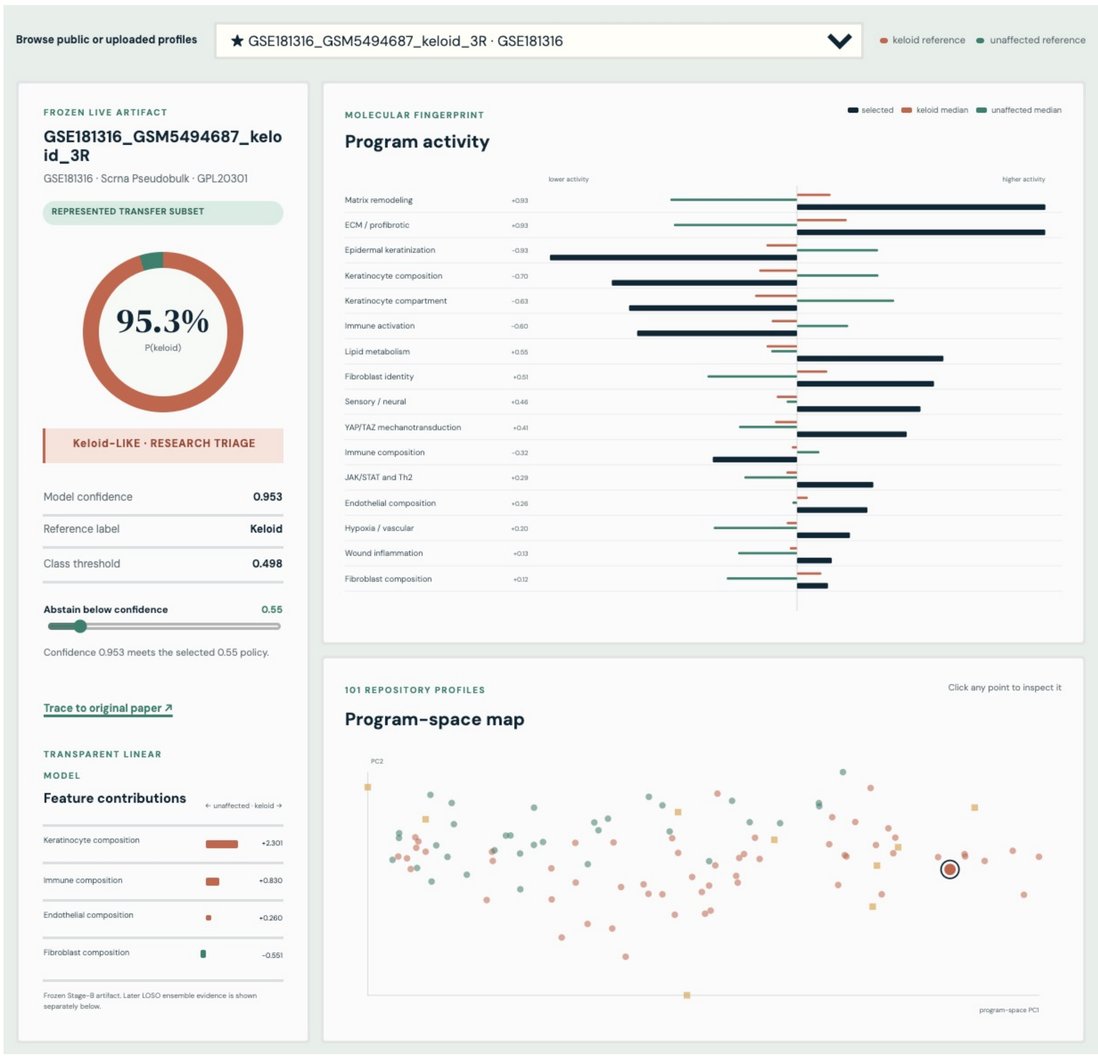
**Algorithm 1** Leakage-safe nested LOSO protocol (selection, prediction, scoring)

**Require:** Studies $\{S_i\}_{i=1}^{n}$; lockboxes $L$; views $\mathcal{V}$; models $\mathcal{M}$; grid $\Theta$; metric $m$; donor map $d$ (opt.).

**for** $s \in [n] \setminus L$ **do**
  $\mathcal{T} \leftarrow [n] \setminus (L \cup \{s\})$, $\mathcal{C} \leftarrow \mathcal{V} \times \mathcal{M}$.
  **for** $C \in \mathcal{C}$ **do**
    **for** $t \in \mathcal{T}$ **do**
      $C^{(-t)} \leftarrow \mathrm{fit}(C, \mathcal{T} \setminus \{t\})$; $p_{C,t}(x) \leftarrow P_{C^{(-t)}}(y{=}1 \mid x)$, $x \in S_t$.
    **end for**
    $p_C \leftarrow \cup_{t \in \mathcal{T}} p_{C,t}$; $\tau_C \leftarrow \arg\max_{\tau \in \Theta} m\big(y_{\mathcal{T}}, \mathbb{I}[p_C \geq \tau]\big)$.
    $\mu_C \leftarrow \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} m\big(y_t, \mathbb{I}[p_{C,t} \geq \tau_C]\big)$.
  **end for**
  $\{C_k, \tau_k\}_{k=1}^{K} \leftarrow \mathrm{TopK}(\mu_C)$; fit $C_k$ on $\mathcal{T}$; $p_k(x) \leftarrow P_{C_k}(y{=}1 \mid x)$, $x \in S_s$.
  $\bar{p}(x) \leftarrow \frac{1}{K} \sum_k p_k(x)$; $\tau \leftarrow \mathrm{median}(\tau_1, \ldots, \tau_K)$; $\hat{y}(x) \leftarrow \mathbb{I}[\bar{p}(x) \geq \tau]$.
  score($S_s$) at sample-grain, or donor-grain if class-pure: $\bar{p}_u \leftarrow \mathrm{mean}_{x:d(x)=u} \bar{p}(x)$.
**end for**

## Results & Live Benchmarks




## Acknowledgements and References

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